

WEBINARS SERIES ON RAINWATER HARVESTING

20th September 2022

Module n°6:

RWH for flood risk management



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Module 6: RWH for flood risk management

Session: Practices and Effectiveness of RWH Systems in Flood Prevention and Mitigation

Date: 20th September 2022 **Time:** 13 h 00 – 14 h 30 (GMT+2)

Register: Practices and effectiveness of rainwater harvesting systems in flood prevention and mitigation

BACKGROUND

Water scarcity is a critical challenge in countries and regions with arid and semi-arid conditions and fast-growing water demands. This is mostly the case in all of the countries across the NENA Region, where food demand and food insecurity continue to grow, widening the water supply-demand gap in most countries. Furthermore, the anticipated impact of climate change in the NENA region is indicative towards an increased intensity of rainstorms and flash floods, while the total amount of annual precipitation rates shows declining trends. During the last 20 years, interest has been renewed in rainwater harvesting (RWH) as one of the most effective climate adaptation strategies to cope with water scarcity, thus reducing the pressure on the limited freshwater resources and increasing rainfed crop productivity.

This webinar series on Rainwater Harvesting aims to provide an overview of RWH applications and illustrates, with country examples, the diversity of approaches. It will try to unlock the potential of rainwater harvesting applications in the NENA region by highlighting the role of rainwater management in overcoming the current challenges in achieving water security. It will also shed the light on the need for innovations in economics, social consensus and supportive laws and regulations to ensure a widespread uptake of RWH technologies in the NENA region.

Main Conveners:

FAO/WEPS-NENA project & Water Scarcity Initiative (WSI) with: ICARDA, ICID and FAO inter-Regional Technical Platform on Water Scarcity (iRTP-WS)

Contributing Partners:

MetaMeta (Netherlands) and the University of Natural Resources and Life Sciences, Vienna (Austria)

Contact Persons:

Domitille Vallée Domitille.Vallee@fao.org

Fahide Si Tahar Fahide.SiTahar@fao.org

Heba Al Hariry Heba.alhariry@fao.org

Simon Tanios Tanios.Simon@fao.org

Languages:

English and French with live translation

Recording:

<https://dgroups.org/fao/waterproductivity/rwh>

ABOUT THE MODULE 6

This module explores the role of RWH in controlling flood risks and using floods productively in rural areas. Several approaches are presented (e.g. road water harvesting, flood spreading) and their effectiveness to control floods is further discussed. Case study elements in the MENA region are included in the presentations.



EXPECTED OUTCOMES FOR THE PARTICIPANTS

The webinar will achieve the following:

- 1 Explore the role of rainwater harvesting in preventing and managing floods;
- 2 Examine the current practices, challenges and opportunities associated with RWH and flood management and utilisation;
- 3 Strengthen the knowledge and awareness of flood management from various RWH perspectives.

SUGGESTED READINGS

- **FAO.** 2010. Guidelines on spate irrigation. FAO irrigation and drainage paper 65, Rome: FAO
<https://floodbased.org/download/182-guidelinesforspateirrigationfao-pdf/>
- **World Bank.** 2021. Green roads for water: guidelines for road infrastructure in support of water management and climate resilience
https://roadsforwater.org/wp-content/uploads/2022/02/WB_Guidelines_final_publication.pdf
- **Haddad, M., Strohmeier, S.M., Nouwakpo, K., Rimawi, O., Weltz, M., Sterk, G.** 2022. Rangeland restoration in Jordan: Restoring vegetation cover by water harvesting measures. International Soil and Water Conservation Research. <https://doi.org/10.1016/j.iswcr.2022.03.001>.
- **Strohmeier, S., Fukai, S., Haddad, M., Al Nsour, M., Mudabber, M., Akimoto, K., Yamamoto, S., Evett, S., Oweis, T.** 2021. Rehabilitation of degraded rangelands in Jordan: The effects of mechanized micro water harvesting on hill-slope scale soil water and vegetation dynamics. Journal of Arid Environments. 185.
<https://doi.org/10.1016/j.jaridenv.2020.104338>.
- **WOCAT** entries: Meso-scale Water Harvesting (Marab):
https://qcat.wocat.net/ar/wocat/technologies/view/technologies_5770/ and Micro-scale Water Harvesting (Vallerani): https://qcat.wocat.net/en/wocat/technologies/view/technologies_5860/



CERTIFICATE OF ACHIEVEMENT

A digital certificate of achievement will be offered by FAO to participants who:

- Attend at least four webinars belonging to Module 1 to Module 4 and five webinars belonging to Module 5 to Module 10;
- Obtain a score of no less than 70 percent in each of two written assessments that include multiple-answer questions and reflect the content of the aforementioned modules.

The first assessment covering the modules 1 to 4 was organized in August 2022. The link to the second assessment, covering the modules 5 to 10, will be sent by email at the end of the year to eligible participants who will be given two weeks to submit their answers. Respondents will have the possibility to make two attempts to answer the assessment questions, and only the highest score will be considered.

Important note: For participants who wish to obtain a certificate of achievement, to be able to track your attendance, you must register and join the webinars using the same e-mail address.

Speakers and moderator



Frank van Steenberghe

has been involved in international policy initiatives such as the Framework for Action Dialogue on Water Governance Guidelines for Spate Irrigation (FAO) and the 'Water for Development' policies for four governments. He was leading the writing team of the Groundwater Governance Vision and Framework for Action – the latter a joint effort of the FAO, UNESCO, World Bank, GEF and IAH. He recently completed the Water Sector Policy for the Islamic Development Bank and Guidelines on Green Roads for Water for the World Bank. He is working with MetaMeta (www.metameta.nl), a social enterprise dedicated to better water management that won two recent environmental awards.



Stefan Strohmeier

is a Senior Researcher at the University of Natural Resources and Life Sciences (BOKU), Institute for Soil Physics and Rural Water Management (SoPhy) in Vienna, Austria. He holds a PhD in Environmental Engineering and Water Management. Prior to his re-engagement at BOKU University in 2022, he spent the past eight years in the dry regions of North Africa and Near East, East- and Sub-Saharan Africa and Central Asia. From 2014 to 2022, Stefan held scientific and project managerial positions at ICARDA, Amman office. From 2020 to 2022, he led ICARDA's research sub-team called "Restoration Initiative on Dryland Ecosystems". Stefan's research interests are on sustainable land and water management towards combatting land degradation. Stefan published dozens of SCI publications and book chapters including written publications and presentations provided at international fora.



Simon Tanios

holds a Master of Engineering Degree in Structure and Public Works, a Master in Cooperation and Development Studies, and a post-graduate Diploma in Data, Economics and Development Policy. He has five years of experience working on infrastructure projects in the WASH sector in Lebanon. Currently, he is a Thematic Networking and Water Governance Analysis Consultant as part of the Water Efficiency, Productivity and Sustainability team at FAO and a Data Analysis and Reporting Consultant for Welthungerhilfe e.V. in Lebanon.

Module 6: RWH for flood risk management

Session: Practices and Effectiveness of RWH Systems in Flood Prevention and Mitigation

Date: Tuesday, 20th September, 2022

Time: 01:00 PM - 02:30 PM (GMT +2)

Time	Activity
01:00 – 01:05	Welcoming Remarks Introduction to the session objectives and speakers By Simon Tanios, Thematic Networking and Water Governance Analysis Consultant, FAO Lebanon
01:05 – 01:30	Presentation on “Non-conventional RWH at scale: water from floods, water from roads” By: Frank van Steenberg, Director at MetaMeta
01:30 – 01:55	Presentation on “RWH for flood-risk management: case study from the Middle East” By: Stefan Martin Strohmeier, Senior Researcher at the University of Natural Resources and Life Sciences, Vienna
01:55 – 02:25	Open Discussions – Q&A's
02:25-02:30	Conclusion & Wrap-up



This activity is implemented within the regional project "Implementing the 2030 Agenda for water efficiency/productivity and water sustainability in NENA countries" under the umbrella of the Water Scarcity Initiative. This regional project is implemented by the Food and Agriculture Organization of the United Nations and funded by the Swedish International Development Cooperation Agency (SIDA).

Food and Agriculture Organization of the United Nations
Regional Office for the Near East and North Africa
www.fao.org/neareast
RNE-WEPS-NENA@fao.org

